

DW03_lfsr_updn

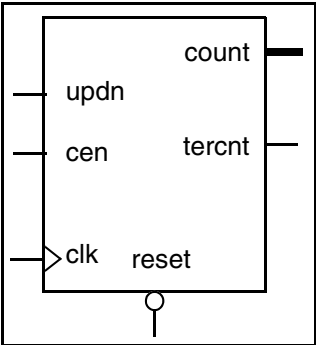
LFSR Up/Down Counter

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Features and Benefits

- High speed, area-efficient
- Pseudo-random sequence generator
- Up/down count control
- Asynchronous reset
- Terminal count flag

Revision History



Description

DW03_lfsr_updn is a programmable word-length counter. DW03_lfsr_updn implements a counter as an LFSR (linear feedback shift register) which also acts as a pseudo-random counter constructed as primitive characteristic polynomials.

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
updn	1 bit	Input	Input high for count up and low for count down
cen	1 bit	Input	Input count enable
clk	1 bit	Input	Clock
reset	1 bit	Input	Asynchronous reset, active low
count	<i>width</i> bits	Output	Output count bus
tercnt	1 bit	Output	Output terminal count

Table 1-2 Parameter Description

Parameter	Values ^a	Description
width	2 to 50	Word length of counter

a. The upper bound of the legal range is a guideline to ensure reasonable compile times.

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
str	Synthesis model	DesignWare

Table 1-4 Simulation Models

Model	Function
DW03.DW03_LFSR_UPDN_CFG_SIM	Design unit name for VHDL simulation
dw/dw03/src/DW03_lfsr_updn_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW03_lfsr_updn.v	Verilog simulation model source code

Table 1-5 Counter Operation Truth Table

reset	updn	cen	Operation
0	X	X	Reset
1	X	0	Standby
1	0	1	Count down
1	1	1	Count up

The shift register is fed back from two or more taps. The number of taps does not increase with a large counter *width*.

An LFSR counter runs faster than a binary counter in synchronous systems because the first bit is calculated and the remaining bits are shifted.

Applications of DW03_lfsr_updn include high-speed counters requiring low area, pseudo-random sequence generation, and built-in self-test (BIST).

Counter Function

The *updn* input port controls the counter direction. When *updn* is high, the counter increments; when *updn* is low, the counter decrements.

When the count enable pin, *cen*, is high, the counter is active. When *cen* is low, the counter is disabled and count remains at the same value.

The *reset*, active low, is an asynchronous reset signal. When *reset* is low, the counter output is "00...00". When *reset* is high, the counter operates normally.

The *count* is the output port, ranging from *width* - 1 to 0. A value of $2^{width-2}$ ("11...11") is an illegal state; therefore, the counter stops at "11...11".

The `tercnt` is an output terminal count signal, active high. The signal `tercnt` goes high for one clock cycle to indicate the different number of clock cycles between starting count to `count_to` value.

Timing Diagrams

The following diagrams show various timing conditions for `DW03_lfsr_updn`.

Figure 1-1 Functional Operation: reset, up_count and count_enable Sequence

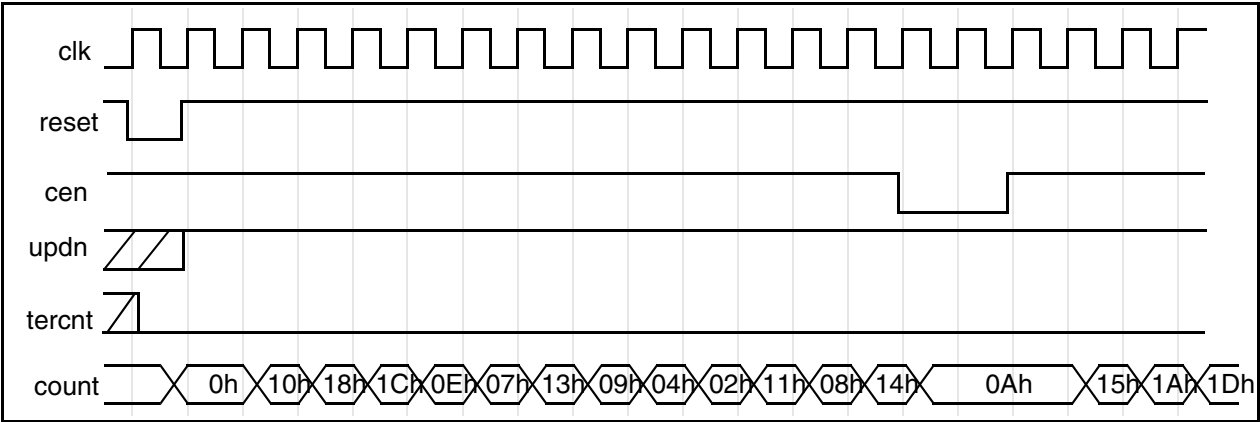


Figure 1-2 Functional Operation 2

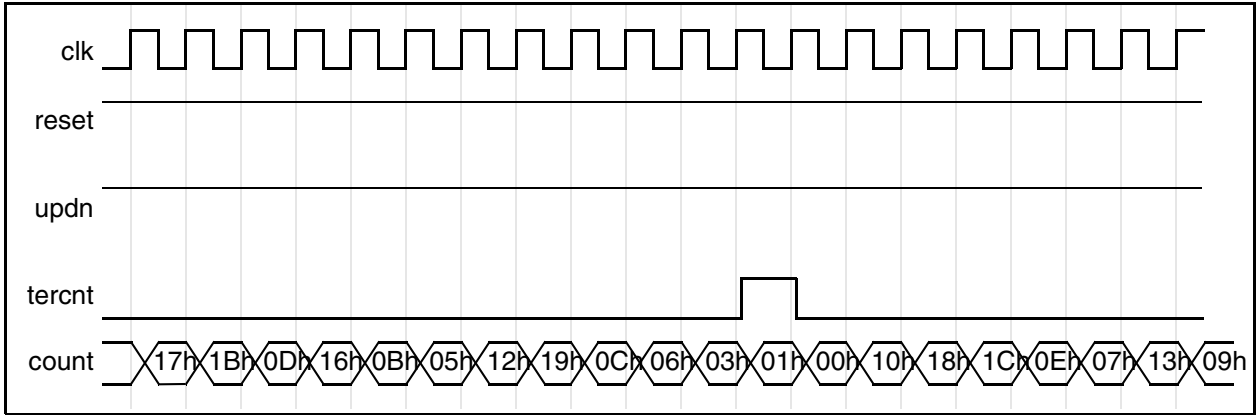


Figure 1-3 Functional Operation: down_count Sequence

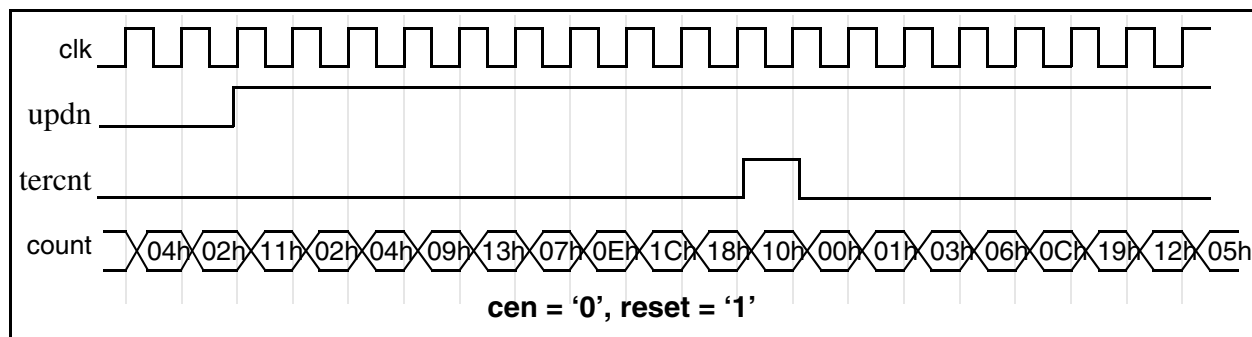


Figure 1-4 Functional Operation: up and down_count Sequence

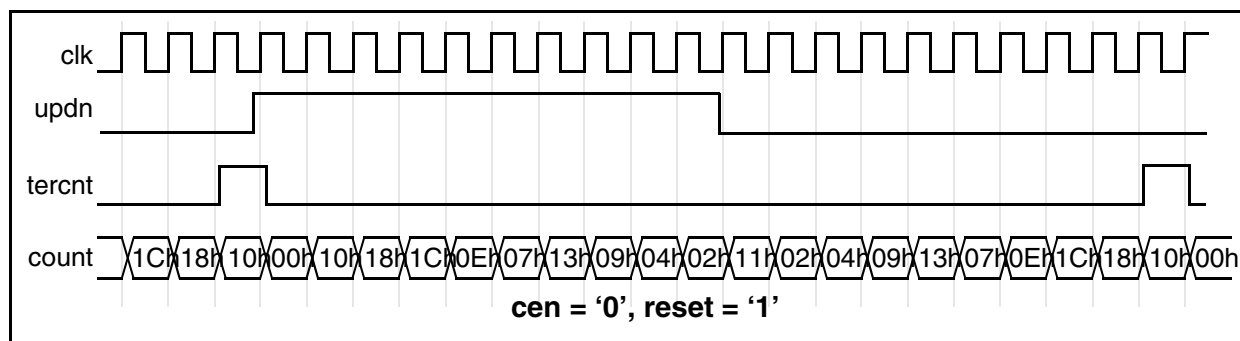
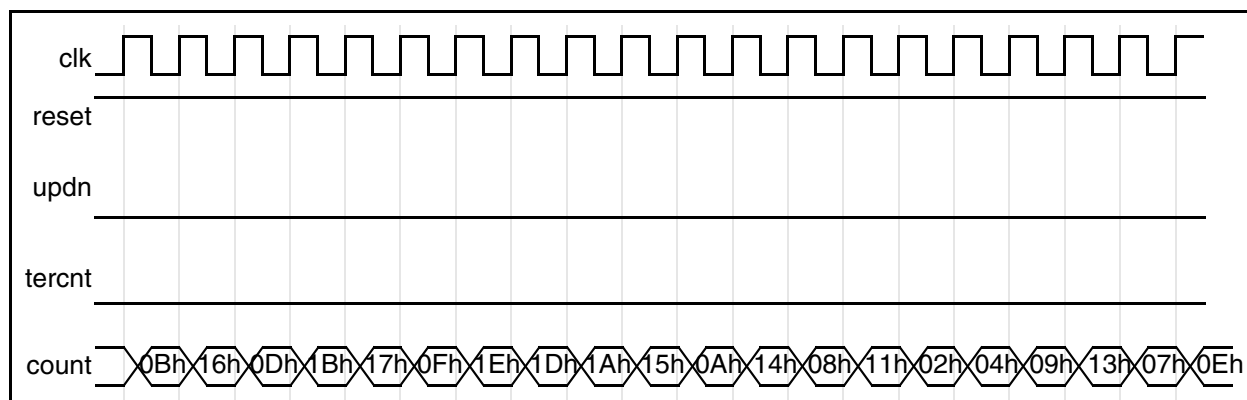


Figure 1-5 Functional Operation 5



Related Topics

- [Logic – Sequential Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_foundation_comp.all;

entity DW03_lfsr_updn_inst is
  generic ( inst_width : INTEGER := 8 );
  port ( inst_updn      : in std_logic;
        inst_cen       : in std_logic;
        inst_clk       : in std_logic;
        inst_reset     : in std_logic;
        count_inst     : out std_logic_vector(inst_width-1 downto 0);
        tercnt_inst    : out std_logic );
end DW03_lfsr_updn_inst;

architecture inst of DW03_lfsr_updn_inst is
begin

  -- Instance of DW03_lfsr_updn
  U1 : DW03_lfsr_updn
    generic map ( width => inst_width )
    port map ( updn => inst_updn, cen => inst_cen, clk => inst_clk,
              reset => inst_reset, count => count_inst,
              tercnt => tercnt_inst );

end inst;

-- pragma translate_off
configuration DW03_lfsr_updn_inst_cfg_inst of DW03_lfsr_updn_inst is
  for inst
    end for; -- inst
end DW03_lfsr_updn_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW03_lfsr_updn_inst( inst_updn, inst_cen, inst_clk,  
                           inst_reset, count_inst, tercnt_inst );  
  
    parameter width = 8;  
  
    input inst_updn;  
    input inst_cen;  
    input inst_clk;  
    input inst_reset;  
    output [width-1 : 0] count_inst;  
    output tercnt_inst;  
  
    // Instance of DW03_lfsr_updn  
    DW03_lfsr_updn #(width)  
        U1 ( .updn(inst_updn), .cen(inst_cen), .clk(inst_clk),  
            .reset(inst_reset), .count(count_inst), .tercnt(tercnt_inst) );  
  
endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
March 2019	DWBB_201903.0	■ Removed minPower designation from this datasheet
January 2019	DWBB_201806.5	■ Updated example in “HDL Usage Through Component Instantiation - VHDL” on page 5 ■ Added this Revision History table and the document links on this page

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